Protecting Location Privacy by Multiquery: A Dynamic Bayesian Game Theoretic Approach

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***Abstract*— When using location-based services (LBSs), a user obtains points-of-interest (PoI) information by providing the LBS platform with his current geo-location. Such a search leads to potential privacy leakage if an adversary has access to his geo-data. Traditional k-anonymity mechanisms instruct a user to bear the overhead to report his current location together with $k-1$ dummy locations to confuse the adversary, which only work well given a large number k. Aware of the common practices that a user is actually flexible in service requirement (e.g., as long as the searched PoIs are within his walking distance), we propose a novel approach to help the user gain location privacy from service flexibility for the challenging case of a small number k. By analyzing the strategic interaction between the user and the adversary in a dynamic Bayesian game, we prove that the user's equilibrium strategy depends on the adversary's capability of accessing geo-data. Take $k = 2$ for example, we find that if the adversary is not likely to access both geo-data, the user should report the two dummy locations at two different directions of his real location, and otherwise at the same direction. Perhaps surprisingly, the user may benefit from the adversary's access to more geo-data. Furthermore, we extend the game-theoretic approach for multi-query and arbitrary user location distributions. Numerical results show that our approach obviously outperforms k-anonymity mechanisms especially under a small number k.**

***Index Terms*— Location-based services, location privacy protection, dynamic game under incomplete information, Bayesian Nash equilibrium.**

I. Introduction

TODAY, many users obtain significant benefits from using location-based services (LBSs) by quickly searching for points-of-interest (PoIs) nearby, such as restaurants, nightclubs, and people for dating [2]. The LBS platform first asks a user to enter his current location as geo-data and then returns the information of target PoIs (if any) within the query range (e.g., walking distance) to the user. During such a query, an adversary may eavesdrop the user's geo-data report or hack into the LBS platform, resulting in severe privacy loss for the user. Such data breach accidents have happened at large scales from time to time. For example, McDonald's delivery app, McDelivery, asks for users' current locations for delivery, but inside this app a leaky Application Program Interface (API) was discovered by a hacker in 2017 to leak over 2.2 million users' private data [3]. Another example is the popular social application Foursquare, where millions of users were involved in data breaches, leaks, hacks, and other data incidents [4]. Hence it is a critical issue to protect the users' privacy on LBS platforms.

To help users protect location privacy while using LBSs, researchers have proposed many mechanisms, such as k-anonymity [5], [6], mix zones [7], dummy generation [8]–[10], and perturbation mechanisms [11]–[13]. The location obfuscation with k-anonymity is a widely-used type of privacy-preserving mechanism, and there are more recent innovations on this (e.g., [5], [6]). Essentially, such techniques mix a user's real location with another $k-1$ dummy locations in his LBS query to create uncertainty among k reportings for the adversary. Similarly, Sen *et al.* in [7] used the mix zone concept to achieve anonymization, within which pseudonyms are exchanged between users. Liu *et al.* in [9] and Wu *et al.* in [10] proposed to generate and report multiple dummy locations to the LBS platform to increase uncertainty for the adversary and thus achieve anonymity, without employing any trusted servers as anonymizer. Under these kinds of mechanisms, the user always receives the PoI information exactly at his real location (as a rigid service requirement), yet incurs overhead and inconvenience for searching $k-1$ extra dummy locations. In the recent perturbation mechanisms [11]–[13], the concept of geo-indistinguishability for location privacy was generalized from the notion of differential privacy [14]. Such mechanisms provide rigorous privacy guarantees, from which we benefit for the privacy definition.

In practice, many users are actually flexible in LBS service requirement. They are not only interested in the PoI information exactly at their real locations but also those PoIs around them [15], which can be obtained by the range query using only a user's reported location and the query range [16]. Accordingly, many LBS platforms recommend users other PoIs within the query range from their reported locations.

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For example, after a user's search using his current geo-data, the popular app booking.com also recommends those hotels less than a query range (1 kilometer (km), 3km and even 5km). Given that such LBS systems are common in practice and provide the service flexibility opportunity, a user no longer has to report his real location for the LBS query. Still, we wonder whether and how dummy generation can help a user best gain location privacy from his service flexibility, by optimally reporting dummy locations. Note that k-anonymity mechanisms only work well given a large number k of dummy locations for enough obfuscation, whereas we aim to design very novel reporting strategy for the user to gain location privacy even in the challenging case of a small number k.

In the literature, there are a few works exploring the service flexibility for privacy preservation. Li and Li in [17] first investigated the tradeoff between privacy and service utility in data publishing. Recent work followed [17] to design the privacy-preserving mechanisms for crowdsensing [18], multi-agent [19] and LBS systems [20]. Several other papers (e.g., [21], [22]) further considered intelligent attacks and analyzed the game-theoretical interaction between a user and an adversary for privacy protection. In particular, Wang and Zhang in [21] took inference attacks from the adversary into consideration, and studied how a user best preserves the location privacy by reporting his location with a certain level of granularity. Shokri *et al.* in [22] aimed to preserve users' privacy under a constrained service requirement. By using the framework of zero-sum Stackelberg games, they proposed an optimal location-privacy preserving algorithm against an adversary implementing an inference algorithm.

These game-theoretic studies, however, limited the user's strategy to LBS query using only $k = 1$ reported location. Furthermore, the existing literature is under complete information and ignores the practical information asymmetry between the user and the adversary. Unlike the existing works, in this paper we explore the good opportunity for the user to jointly use more than one dummy location for better performance by balancing the service and privacy gains. Although the multi-query with service flexibility offers the user more choices, a potential risk here is that a user's real location may be inferred more easily when the intelligent adversary accesses both/all geo-data.

Under such motivations and concerns, we address the gap of exploring the potentials of LBS service flexibility to achieve better location privacy, by strategically using multiple queries. This yields the potential benefits compared with the existing privacy-preserving techniques with rigid service requirement. The key novelty and main contributions of this paper are summarized as follows.

- *Novel approach to preserve location privacy by using LBS service flexibility*: To the best of our knowledge, this is the first work that theoretically studies how to strategically use a user's service flexibility to preserve his location privacy, by jointly designing k queries (geo-data) for LBS. Compared with the traditional single-query with $k = 1$ analyzed in the literature, we prove that our dual-query approach with $k = 2$ leads to a greater user payoff. Based on realistic data, we show that the user payoff increase with dual-query is between 26.7% and 37.8%.
- *Dynamic Bayesian game theoretic modelling and robust analysis*: To ensure reliable performance, we consider the worst-case scenario with asymmetric information for the user to make a robust defence against an adversary. In the challenging but practical scenario, the user does not know how many geo-data are accessed by the adversary, while the adversary has the strongest knowledge of the user's payoff function and performs the optimal inference attack. This is the first work to formulate the strategic interplay as a dynamic Bayesian game with a theoretic equilibrium.
- *The impact of our dual-query strategy using service flexibility*: Our results show that the user's reporting strategy depends on the possibility of the adversary to access one or both/all geo-data at the Bayesian Nash equilibrium (BNE). Take $k = 2$ for example, as the chance for the adversary to access both geo-data increases, the user's reporting strategies will change from two different directions to the same direction of his real location. Perhaps surprisingly, our approach may allow the user to benefit from the adversary's access to more geo-data, as the user can strategically report locations to make it more difficult for the adversary to infer the real location.
- *Obvious performance gain over k-anonymity mechanisms*: When extending our approach to arbitrary k dummy locations, the intelligent adversary may better infer the user's location from more geo-data and the user should proactively decide reports to mislead the adversary. For the typical case of $k = 3$, we prove that the user should design the three dummy locations to form an isosceles triangle. The user's final payoff increases with the number of queries k, but with a diminishing return. Even if the user's location is arbitrarily distributed and the adversary has such background information, our strategic approach using service flexibility obviously outperforms the modified k-anonymity mechanisms especially under a small number k.

The rest of the paper is organized as follows. Section II introduces the system model. Section III analyzes the adversary's best inference attack strategy of the user's real location when it accesses part or all of geo-data. Sections IV and V study the user's equilibrium reporting strategy for a representative case of $k = 2$ in one-dimensional (1D) and two-dimensional (2D) cases, respectively. Section VI extends the approach to the general k scenario and compares its performance with the k-anonymity mechanism. Section VII extends the approach with a general location distribution and consider the adversary's background knowledge. Finally, we conclude this work in Section VIII. Due to space limit, we put the detailed proofs in the on-line technical report [23].

II. System Model

Figure 1 describes the interaction between an LBS user and an adversary. The LBS platform will return the PoI information

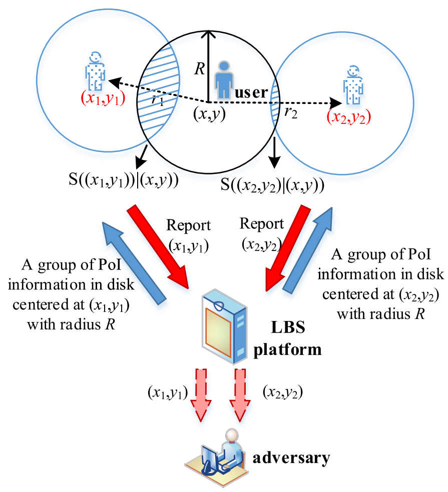


Fig. 1. System model for privacy protection, where the user with real location (x, y) reports (x_1, y_1) and (x_2, y_2) on the LBS platform. The platform provides the user with a group of nearby PoI information in the circular disks centered at (x_1, y_1) and (x_2, y_2) with radius R, respectively. After the user's geo-data reports, the adversary has probability η to access both geo-data and $1 - \eta$ to access only one.

based on the user's reported location data, and it is not a strategic decision maker in this model. Given the traditional k-anonymity mechanisms do not work well under a small number k of dummy locations, we focus on the challenging case of a small k to help the user gain location privacy from the service flexibility. Without much loss of generality, we look at the case of dual-query with $k = 2$ in this section, and still extend our theoretical model and analysis to the more general multi-query case with arbitrary k's in Section VI.

More specifically, the user is flexible in requesting the information about nearby PoIs within a radius R from his real geo-location (x, y) in the two-dimensional (2D) ground plane, e.g., nightclubs or restaurants within the walking distance. As shown in Fig. 1, the user first reports two dummy locations/geo-data (x_1, y_1) and (x_2, y_2) to search the PoI information around, both of which may be different from his real location (x, y) for the purpose of privacy preservation. Then the platform returns to the user nearby PoIs within two different disks of the same radius R, centered at the search locations (x_1, y_1) or (x_2, y_2), respectively.

The adversary may have access to one or both geo-data of (x_1, y_1) and (x_2, y_2) to infer the user's real location. Without loss of generality, we do not consider the trivial case when the adversary cannot access any geo-data hence the user suffers from no privacy loss. It should also be noted that it is not guaranteed that the adversary has access to both geo-data, as the user may query two competitive LBS platforms and use VPN to hide his identity [24]. To hide the real location from such attacks, the user needs to strategically choose (x_1, y_1) and (x_2, y_2) on the platform and balance the LBS service and privacy gains.

Next, we will introduce the adversary's inference strategy of user's real location in Section II-A, then model user's payoff including flexible service and privacy gains in Section II-B, and finally model the two-stage Bayesian game under asymmetric information in Section II-C. Table I summarizes some key notations in this paper.

TABLE I

KEY NOTATIONS USED IN THE PAPER AND THEIR PHYSICAL MEANINGS

Notation	Physical Meaning
(x, y)	User's real location.
(x_i, y_i)	User's reported dummy locations for LBS.
R	User's query range to tell service flexibility.
$d(A, B)$	Euclidean distance between any points A and B.
$(\hat{x}^{BNE}, \hat{y}^{BNE})$	Adversary's guess using both geo-data at the BNE.
$(\hat{x}_i^{BNE}, \hat{y}_i^{BNE})$	Adversary's guess with only (x_i, y_i) at the BNE.
r_i	User's reporting distances between (x_i, y_i) and (x, y).
λ	PoIs' spatial density in LBS.
$\bar{N}$	Mean number of PoIs within a service coverage area.
$f(\cdot)$	User's flexible service gain function.
η	Adversary's access probability to two geo-data.
θ	User's reporting angle $\langle (x_1, y_1), (x, y), (x_2, y_2) \rangle$.
r_i^*	User's same-sided reporting distance at the BNE.
r_i^{**}	User's two-sided reporting distance at the BNE.
r_i^{BNE}	User's reporting distance at the BNE.
x_i^{BNE}	User's reporting location at the BNE.
k	Number of LBS queries.

A. Adversary's Inference Attack of User's Real Location

Depending on the number of geo-data that the adversary can access, it will infer the user's real location (x, y) from user's reported geo-data (x_1, y_1), or (x_2, y_2), or both. We consider an intelligent attack, where the adversary's objective is to minimize the inference error (i.e., expected Euclidean distance from the guessed location to user's real location).

First, we consider the case where the adversary can only access one reported location, say geo-data (x_i, y_i) $(i \in \{1, 2\})$. Then the adversary will minimize the expected Euclidean distance from the guessed location to the user's real location, i.e., solving the following location estimation problem:

$$\left(\hat{x}_i^{BNE}(x_i, y_i), \hat{y}_i^{BNE}(x_i, y_i)\right) = \arg\min_{(x_i', y_i')} \mathbb{E}_{(X,Y)} d\left((x_i', y_i'), (X, Y)\right), \quad (1)$$

where (X, Y) are 2D random variables denoting the possible locations for the real location (x, y) given the user's reported geo-data (x_i, y_i). The superscript "BNE" above corresponds to "Bayesian Nash Equilibrium" as the stable outcome of our Bayesian game. In (1), we let $d((x_i', y_i'), (X, Y))$ denote the Euclidean distance between two points (x_i', y_i') and (X, Y) on the 2D ground plane, i.e.,

$$d((x_i', y_i'), (X, Y)) = \sqrt{(x_i' - X)^2 + (y_i' - Y)^2}.$$

The **best inference** $(\hat{x}_i^{BNE}, \hat{y}_i^{BNE})$ **for the single-data hacking case** is a function of (x_i, y_i).

Next we consider the case where the adversary can access both reported locations. It will infer $(\hat{x}^{BNE}, \hat{y}^{BNE})$ from both (x_1, y_1) and (x_2, y_2), i.e.,

$$\left(\hat{x}^{BNE}((x_1, y_1), (x_2, y_2)), \hat{y}^{BNE}((x_1, y_1), (x_2, y_2))\right) = \arg\min_{(x', y')} \mathbb{E}_{(X,Y)} d\left((x', y'), (X, Y)\right), \quad (2)$$

where (X, Y) are 2D random variables denoting the possible locations for (x, y) given two geo-data (x_1, y_1) and (x_2, y_2).

The **best inference** $(\hat{x}^{BNE}, \hat{y}^{BNE})$ **for the two-data hacking case** depends on (x_1, y_1) and (x_2, y_2) jointly.

Later in Section III, we will analyze the adversary's best inference attack strategy in more details, i.e., how (1) and (2) would be computed against the user's location reporting.

B. User's Payoff With Flexible Service and Privacy Gains

We first introduce the user's strategic multi-query model, then define the user's flexible service and privacy gains. Based on these, we formulate the user's expected payoff that involves both his service and privacy gains.

1) User's Strategic Location Reporting: In general, the user may report dummy locations different from his real location to the LBS platform. As illustrated in Fig. 1, the user strategically reports two geo-data (x_i, y_i) $(i \in \{1, 2\})$ for the LBS. The **reporting distance** r_i measures the Euclidean distance between the reported location and the real one, i.e.,

$$r_i = d\left((x_i, y_i), (x, y)\right) = \sqrt{(x_i - x)^2 + (y_i - y)^2}. \quad (3)$$

Given the real location (x, y), we denote $S((x_i, y_i)|(x, y))$ as the service coverage area between the two disks centered at (x_i, y_i) and (x, y) in Fig. 1. Though flexible, the user only finds the PoIs less than walking distance R useful, including those PoIs in overlap regions $S((x_1, y_1)|(x, y))$ and $S((x_2, y_2)|(x, y))$ in the 2D ground plane. The **total service coverage area** by querying the LBS platform is the union set

$$S((x_1, y_1), (x_2, y_2)) = S((x_1, y_1)|(x, y)) \cup S((x_2, y_2)|(x, y)), \quad (4)$$

by combining the useful service coverages returned by both queries.

2) User's Service Gain: The user's service gain measures his benefit from finding PoIs on the platform, which depends on the service coverage and PoI density as explained in the following.

First, we consider that the locations of PoIs follow a 2D Poisson point process (PPP) with a **spatial density** λ and thus a PoI occurs spatially with equal probabilities at each point in the considered region. Note that PPP is widely used in the literature to model the spatial distribution of facility nodes on the ground plane (e.g., [25], [26]). Given that PoIs are uniformly distributed with uniform density λ, we use a random variable $N(\lambda, S)$ to denote **the total number of PoIs** in the service coverage area S. First we have the probability mass function (PMF) of having N PoIs as

$$\Pr(N = n) = \frac{(\lambda S)^n}{n!} \exp\left(-\lambda S\right), \quad n = 0, 1, \cdots.$$

Then we can obtain the mean of $N(\lambda, S)$ is $\bar{N} = \lambda S$, which increases with the service coverage area S and the PoI density λ. Note that the PPP model helps us to easily use the service coverage area S per query to indicate the expected number of PoIs in the service coverage area. Other non-PPP models is also applicable, though it will become more involved to compute the mean PoI number across different regions of different densities.

Next we define the user's **flexible service gain** in a general way with respect to the mean number $\bar{N}$ of PoIs in the total service coverage area S in (4), which conforms to the law of diminishing returns in the literature [27], [28].

Definition 1 (User's Flexible Service Gain Function): The user's service gain function $f(\bar{N})$ is an increasing and concave function in the mean number $\bar{N} = \lambda S$ of the searched PoIs. The function satisfies $f(0) = 0$, $f(\bar{N}) \geq 0$, $f'(\bar{N}) > 0$ and $f''(\bar{N}) < 0$ for $\bar{N} \geq 0$.

Based on Definition 1, $f(\bar{N})$ can take the forms of $f(\bar{N}) = 1 - e^{-\bar{N}}$, $f(\bar{N}) = \log(\bar{N})$, $f(\bar{N}) = \bar{N}^{\alpha}, \alpha \in (0, 1)$, etc (see Table I in [28]). All the functions f are directly related to the mean number $\bar{N}$ of searched PoIs. Further to say, it should depend on the user's real location, reported location(s), the number of his queries and the PoI density λ.

3) User's Privacy Gain: The user's privacy gain by strategic location reporting depends on whether the adversary can access one or both geo-data, as well as the adversary's attack strategy of inferring the user's location.

According to [29], (location) privacy gain corresponds to the ability for the user to prevent the adversary from guessing his real location (x, y). Here we define the **privacy gain** as the minimum Euclidean distance from the guessed location to the user's real location, which is given as

$$\text{Privacy gain} = \begin{cases} \min_{i=1,2} d\left((\hat{x}_i^{BNE}, \hat{y}_i^{BNE}), (x, y)\right), \\ \text{if adversary accesses one geo-data,} \\ d\left((\hat{x}^{BNE}, \hat{y}^{BNE}), (x, y)\right), \\ \text{if adversary accesses both geo-data.} \end{cases} \quad (5)$$

In the first case, the minimum operation implies the fact that the user does not know which geo-data that the adversary can access, hence the user will consider the worst case [30]. Note that the privacy definition is not constrained in the given form in (5). In the field of location-based systems, the concept of differential privacy [14] and geo-indistinguishability [11] provide rigorous mathematical privacy guarantees, and they also apply to our model.

4) User's Payoff: In reality, the user does not know whether the adversary accesses one or both geo-data. Hence we assume the user holds the belief that the adversary accesses both geo-data with an **access probability** η and one geo-data with a probability $1 - \eta$. The user needs to decide the strategic reporting of dummy locations (x_1, y_1) and (x_2, y_2) to achieve the best tradeoff between the flexible service gain $f(\cdot)$ in Definition 1 and case-by-case privacy gain in (5). Given the service flexibility, the user can sacrifice the service gain to better preserve the location privacy. As a product between these two terms, Equation (6), as shown at the bottom of the next page, summarizes the **overall expected payoff** of the user. A special (benchmark) case is when the user does not use any LBS query at all, in which case his privacy gain is maximum yet the service gain is zero, resulting in an overall zero payoff. This motivates the choice of the product form in terms of service gain in Definition 1 and privacy gain (5) when considering the expected payoff in (6).

C. Bayesian Game Formulation

Under the information asymmetry, the user only knows whether the adversary accesses both geo-data in a probabilistic manner. This motivates us to propose a Bayesian game theoretic approach for the user against the strategic adversary. Formally, we model the strategic interaction between the user and the adversary as a two-stage Bayesian game below:

- Players: The user and the adversary.
- Information type: The adversary with the strongest knowledge does not know the user's real geo-location (x, y), but it knows the user's expected payoff function in (6) with $f(\cdot)$ introduced in Definition 1. The user knows the adversary's probability of accessing both geo-data η, but does not know exactly which geo-data that the adversary will access.
- Strategies: The game is played dynamically in two stages. In Stage I, the user makes a deterministic strategy on how to report two locations (x_1, y_1) and (x_2, y_2) on the platform, given his real location (x, y). In Stage II, the adversary infers the user's location as $(\hat{x}_i, \hat{y}_i)$ in (1) or $(\hat{x}, \hat{y})$ in (2), depending on whether it accesses one or both geo-data.
- Objectives: The user's objective is to maximize the expected payoff in (6). The adversary's objective is to minimize the expected distance from the guessed user location to the real user location as in (1) or (2).

The Stakelberg game formulation between the user and the adversary models the interaction as dynamically played, where the user makes the move of reporting first, taking the adversary's inference into consideration. Then the adversary moves to make the optimal attack. Next we will analyze the two-stage game through backward induction. We will first analyze the adversary's best inference strategy in Stage II, and then the users' best reporting strategy in Stage I by predicting and taking the adversary's best inference into consideration.

III. Analysis of Stage II: Predicting the Adversary's Best Inference Attack

In this section, we analyze the adversary's best inference attack strategy in Stage II, under any given user location reporting strategy in Stage I. The user's two reported dummy locations (x_1, y_1) and (x_2, y_2) can have different Euclidean distances r_1 and r_2 to his real location (x, y). Without loss of generality, we denote the smaller reporting distance between two queries as $r_{\min} = \min(r_1, r_2)$, and the larger reporting distance as $r_{\max}$. When the reporting distances of both geo-data are the same, we have $r_{\min} = r_{\max}$.

Notice that in practice, the user can hardly know how much background information the adversary knows (e.g., from hacking its registered profile and historical data under the LBS platform [3], [4]). If the platform is totally compromised by the adversary, the adversary can infer the user's type or payoff function. To ensure reliable performance, our approach is based on the worst-case scenario to provide robust protection against any inference attack. That is, the adversary has the strongest knowledge about the user's payoff function and performs the optimal attack. Such robustness guarantees, have been commonly used in the literature by formulations such as max-min problems [31]–[34]. With the knowledge of the user's payoff function, the adversary can reverse engineer and infer the user's location reporting strategy, i.e., $r_{\min}$ and $r_{\max}$. Yet, we will prove that the adversary's best inference is actually independent of the specific values of $r_{\min}$ and $r_{\max}$.

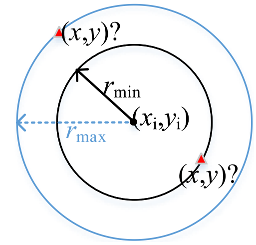


Fig. 2. When the adversary only has access to (x_i, y_i), all possible location candidates for (x, y) appear on the two circles, and are either distance $r_{\min}$ or $r_{\max}$ away from the circles' center (x_i, y_i).

Notice that the user does not have a preference between the two queries due to PoIs' uniform distribution and the adversary cannot distinguish the accessed geo-data. From the perspective of the adversary, it is then equally likely for either accessed geo-data to be with a distance $r_{\min}$ or $r_{\max}$ away from (x, y). To make the best inference attack, the adversary needs to first find out all possible candidates for the real location (x, y) on the 2D ground plane, and then infer one location solution that minimizes the expected distance to the real location among all the candidates.

A. Adversary's Best Inference $(\hat{x}_i, \hat{y}_i)$ From Geo-Data (x_i, y_i) Only

When the adversary can only access the user's one reported dummy location (x_i, y_i) $(i \in \{1, 2\})$, it aims to solve Problem (1) to infer (x, y). It believes that (x, y) is at either the smaller distance $r_{\min}$ or the larger distance $r_{\max}$ away from (x_i, y_i). Thus, all possible candidates for the user's real location (x, y) are equally likely to appear at any points on the two circles centered at (x_i, y_i) with radius $r_{\min}$ and $r_{\max}$, as shown in Fig. 2. As (x_i, y_i) is located at the center of all the possible candidates for (x, y), it has the minimum average distance to all these candidates on either one of the two circles. Therefore, (x_i, y_i) is the unique optimal solution to Problem (1), which does not depend on the specific values of $r_{\min}$ and $r_{\max}$.

Proposition 1: When the adversary can only access the user's one geo-data report (x_i, y_i), its best inference attack

$$U((x_1, y_1), (x_2, y_2)|(x, y)) = \left((1-\eta)\min_{i=1,2} d\left((\hat{x}_i^{BNE}, \hat{y}_i^{BNE}), (x, y)\right) + \eta d\left((\hat{x}^{BNE}, \hat{y}^{BNE}), (x, y)\right)\right) f\left(\lambda S((x_1, y_1), (x_2, y_2))\right). \quad (6)$$

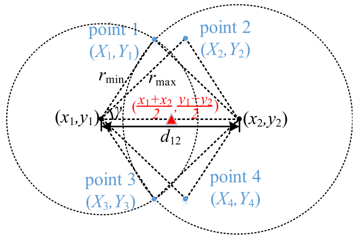


Fig. 3. When the adversary accesses both (x_1, y_1) and (x_2, y_2), there are four candidates for (x, y) to appear at the intersection points of the two circles centered at the geo-data (x_1, y_1) and (x_2, y_2) with radius $r_{\min}$ or $r_{\max}$.

at the BNE is the same as the geo-data report, i.e., $(\hat{x}_i^{BNE}, \hat{y}_i^{BNE}) = (x_i, y_i)$.

Even if the adversary knows the user's reporting distances $r_{\min}$ and $r_{\max}$, it has to fully count on the user's geo-data report since the user still has the freedom to misreport his (x_i, y_i) in any direction from real (x, y) in the 2D plane.

B. Adversary's Best Inference $(\hat{x}, \hat{y})$ From Both Geo-Data

Now we turn to the case that the adversary can access both geo-data (x_1, y_1) and (x_2, y_2), and it aims to solve Problem (2). Here, the adversary manages to associate the eavesdropped geo-data information in anonymized LBS query datasets to individual identities [35].

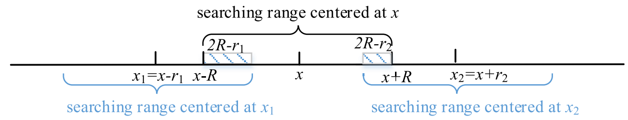
Note that the real location (x, y) is equally likely to be a distance $r_{\min}$ or $r_{\max}$ away from any of the two reported dummy locations. Based on this, Fig. 3 shows the four possible candidates for the real location, each with probability $1/4$, given any $r_{\min}$ and $r_{\max}$ combination. This allows the adversary to better estimate the possible values of (x, y) from the four possible choices. As shown in Fig. 3, the mean of (x_1, y_1) and (x_2, y_2) represented by the red point has the minimum expected distance to the four possible points, hence is the unique optimal solution to Problem (2).

Proposition 2: When the adversary can access the two geo-data (x_1, y_1) and (x_2, y_2), its best inference attack at the BNE is the middle point of the user's two reported dummy locations, i.e., $(\hat{x}^{BNE}, \hat{y}^{BNE}) = (\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$.

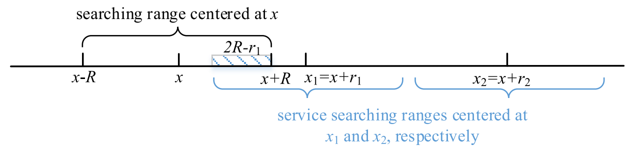
Proposition 2 tells that the adversary's dominant strategy is to use and balance both geo-data to best infer the user's real location. The adversary's equilibrium strategy only relies on the user's reported geo-data and does not depend on the specific reporting strategy $r_{\min}$ or $r_{\max}$.

IV. User's Equilibrium Location Reporting in a 1D Line Scenario

Given the best inference response of the adversary in Stage II, now we study the user's equilibrium reporting strategy with $k = 2$ in Stage I, without knowing whether the adversary can access only one or both geo-data. In this section, for the ease of exposition, we will consider the one-dimensional (1D) case for the user's real and reported locations. Such a model is practical when the user's location is restricted to a long road, such as along a highway. In Section V, we will consider the model in a general two-dimensional (2D) plane scenario. The analysis there will be more involved, yet many key insights derived from the 1D scenario still hold in the 2D scenario.



(a) When reporting x to $x_1 = x - r_1$ and $x_2 = x + r_2$ on the two sides, the user gains both service coverages $2R - r_1$ and $2R - r_2$, respectively.



(b) When reporting x to $x_1 = x + r_1$ and $x_2 = x + r_2$ on the same side, the user only gains service coverage $2R - r_1$ from x_1 but potential privacy gain from x_2.

Fig. 4. User's location searching with radius R in a one-dimensional line.

As shown in Fig. 4, the user with real location x in the 1D line strategically reports (x_1, x_2) on the platform. Let $r_1 = |x_1 - x|$ and $r_2 = |x_2 - x|$ denote the reporting distances. The service coverage $S(x_1, x_2)$ for computing the service gain now only depends on r_1 and r_2. Without loss of generality, we assume $r_1 \leq r_2$. We rewrite $S(x_1, x_2) = S(r_1, r_2)$ for ease of reading here. In Propositions 1 and 2 in Section III, we present the adversary's equilibrium inference strategy in a more general 2D scenario, and the results can be proved to hold for the 1D case as well. Hence, we have $\hat{x}_i^{BNE} = x_i$ for the single-data hacking case and $\hat{x}^{BNE} = \frac{x_1+x_2}{2}$ for the two-data hacking case similarly. The user's expected payoff function in (6) in the 1D scenario is simplified to[1]

$$\begin{aligned} &U(x_1, x_2|x) \\ &= ((1-\eta) \min_{i=1,2} d(\hat{x}_i^{BNE}, x) + \eta d(\hat{x}^{BNE}, x)) f(\lambda S(x_1, x_2)) \\ &= \left((1-\eta) \min_{i=1,2} r_i + \eta \left|\frac{x_1+x_2}{2} - x\right|\right) f(\lambda S(r_1, r_2)), \qquad (7) \end{aligned}$$

which can be rewritten as $U(r_1, r_2)$, given the part $|\frac{x_1+x_2}{2} - x|$ can be written as functions of r_1 and r_2 case-by-case (see the discussion in Section IV-B.1 and Section IV-B.2).

A. Bayesian Equilibrium Analysis With $\eta = 0$

We first consider a special case that the adversary can only access one geo-data, i.e., $\eta = 0$. In this case, the user's expected payoff is simplified from (7) to

$$U(r_1, r_2) = \left(\min_{i=1,2} r_i\right) f(\lambda S(r_1, r_2)). \qquad (8)$$

In the 1D scenario, the user has two choices to report the two locations: either x_1 and x_2 are on the two opposite sides of the real location x as in Fig. 4(a) or on the same side as in Fig. 4(b). For any $r_1, r_2 \geq 0$, we show that the user will always

[1] As the user cannot report a dummy location arbitrarily far away from x to confuse the adversary [36], we limit the reporting distance by L to avoid the trivial case: the user takes advantage of the adversary's mean inference strategy in Proposition 2 to obtain a huge payoff, by reporting a far enough x_2 to obtain a huge privacy gain while keeping x_1 close enough to x for a positive service coverage.

choose the reporting strategy as Fig. 4(a) when the adversary can only access one geo-data.

Lemma 1: If the adversary can only access one geo-data ($\eta = 0$), for any reporting distances r_1 and r_2, the user will report locations x_1 and x_2 on the two opposite sides of the real location x, i.e., $(x_1 - x)(x_2 - x) \leq 0$.

Reporting x_1 and x_2 on the same side of x only brings in service coverage $S(r_1, r_2) = 2R - r_1$ in Fig. 4(b), while reporting on the two opposite sides brings in larger service coverage $S(r_1, r_2) = 4R - r_1 - r_2$ in Fig. 4(a). Thus, given the same privacy gain determined by r_1, the user always prefers to report on the two sides of x to yield greater service coverage. Based on Lemma 1, the following proposition gives the equilibrium reporting distances when strategically reporting geo-data on the two sides of the real location.

Proposition 3: If the adversary can only access one geo-data ($\eta = 0$), the user will strategically report with the same distances $r_1 = r_2 = r^{**} = \min(r^{\dagger\dagger}, L)$, where $r^{\dagger\dagger}$ is the unique solution to

$$f(\lambda(4R - 2r)) - 2\lambda r f'(\lambda(4R - 2r)) = 0. \tag{9}$$

That is, at the BNE, the user will report on the two sides of the real location x with $(x_1 = x \mp r^{**}, x_2 = x \pm r^{**})$.

Given $\eta = 0$, reporting with the same distances on the two sides does not help the adversary infer the user's real location but at least doubles the service coverage. The result shows that the user will prefer to strategically report to two symmetric locations on two sides of x.

B. Bayesian Equilibrium Analysis With $0 \leq \eta \leq 1$

Similar to Section IV-A, given $0 \leq \eta \leq 1$, the user has two choices to report geo-data: two-sided or same-sided. We will analyze each case separately and compare the user's corresponding payoffs, to decide the user's equilibrium reporting strategy. We will show that depending on the value of η, it is no longer always optimal to report the locations on the opposite sides.

1) How to Strategically Report x_1 and x_2 on Two Opposite Sides of x: First, let us consider the case where the user is constrained to report two locations on the two sides of his real location as shown in Fig. 4(a).

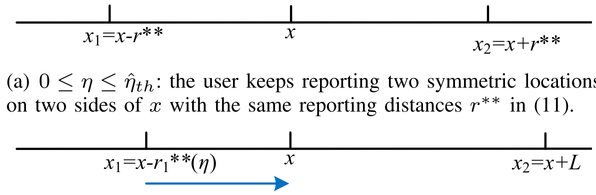
Since the adversary's inference error $d\left(\hat{x}^{BNE}, x\right)$ is $|\frac{x_1+x_2}{2} - x| = \frac{r_2 - r_1}{2}$ with a probability η and is $\min(r_1, r_2) = r_1$ with a probability $1 - \eta$ according to Propositions 1 and 2, respectively, the user's expected payoff in (7) is simplified to

$$\begin{aligned} U_2(r_1, r_2) &= \left((1-\eta)r_1 + \eta\frac{r_2 - r_1}{2}\right) f(\lambda S(r_1, r_2)) \\ &= \left((1 - \frac{3}{2}\eta)r_1 + \frac{\eta}{2}r_2\right) f(\lambda S(r_1, r_2)), \end{aligned} \tag{10}$$

where $S(r_1, r_2)$ is complicated and should be discussed case by case. Though (10) is not a concave function of r_1, we analyze this non-convex problem over different cases of (r_1, r_2) and manage to find the optimum below.

Proposition 4 (User's Best Two-Sided Reporting Strategy): Assume the user reports x_1 and x_2 on two sides of the real location x (i.e., $(x_1 - x)(x_2 - x) \leq 0$), the user's equilibrium reporting strategy on two sides are $(x_1 = x \mp r_1^{**}(\eta), x_2 = x \pm r_2^{**}(\eta))$. There exists a unique threshold $\hat{\eta}_{th} \in (0, 1)$ for the user to decide the same or different reporting distances $(r_1^{**}(\eta), r_2^{**}(\eta))$, as illustrated in Fig. 5.

- *If $0 \leq \eta \leq \hat{\eta}_{th}$,*
$$r_1^{**}(\eta) = r_2^{**}(\eta) = r^{**} = \min\left(r^{\dagger\dagger}, L\right), \tag{11}$$
where $r^{\dagger\dagger}$ is the unique solution to (9) and is the same as the case of $\eta = 0$.
- *If $\eta > \hat{\eta}_{th}$, then $r_2^{**}(\eta) = L$ and $r_1^{**}(\eta)$ is a non-increasing function from $\eta = \hat{\eta}_{th}$ to $\eta = 1$.*[2] *Particularly, when $\eta = 1$, $r_1^{**}(\eta) = 0$.*



(a) $0 \leq \eta \leq \hat{\eta}_{th}$: the user keeps reporting two symmetric locations on two sides of x with the same reporting distances r^{**} in (11).

(b) $\hat{\eta}_{th} < \eta \leq 1$: x_2 keeps as $x + L$, while x_1 moves closer to x as η increases.

Fig. 5. Illustration of the user's best two-sided reporting of x_1 and x_2 as η increases.

Notice that the symmetric reporting distances in (11) for a small value η are consistent with (9) for $\eta = 0$, which is also illustrated in Fig. 5(a). If η is small, the user feels safe to use symmetric $r_1^{**}(\eta) = r_2^{**}(\eta)$ on the two opposite sides of x without worrying that the adversary will use mean location inference to identify his real location. If η is large, the adversary is more likely to take the mean inference strategy. The user will choose $x_1 = x - r_1^{**}(\eta)$ for the service gain, while using the largest possible value $x_2 = x + L$ on the other side for the privacy gain, as illustrated in Fig. 5(b). Finally, when $\eta = 1$, the user will place $(x_1 = x, x_2 = x + L)$, which is the extreme case of reporting on both sides.

2) How to Strategically Report x_1 and x_2 on the Same Side of x: Next we consider the case where the user is constrained to report two locations on the same right-hand side of the real location x as shown in Fig. 4(b).

Under the assumption of $r_1 \leq r_2$, the service coverage is only determined by r_1 with $S(r_1, r_2) = 2R - r_1$. The privacy gain $d\left(\hat{x}^{BNE}, x\right) = |\frac{x_1+x_2}{2} - x|$ equals $\frac{r_1+r_2}{2}$ with a probability η and equals r_1 with a probability $1 - \eta$, then the user's expected payoff in (7) is simplified to a function of r_1 only:

$$\begin{aligned} U_1(r_1) &= \left((1-\eta)r_1 + \eta\frac{r_1 + r_2}{2}\right) f(\lambda S(r_1)) \\ &= \left((1 - \frac{\eta}{2})r_1 + \frac{\eta}{2}L\right) f(\lambda(2R - r_1)), \end{aligned} \tag{12}$$

where $r_2 = L$ as U_1 increases with r_2. Since (12) is a concave function of r_1, by checking the first-order condition we obtain the equilibrium reporting distances to maximize (12).

Proposition 5 (User's Best Same-Sided Reporting Strategy): Assume the user reports on the same side of the real location,

[2]In our technical report [23], we provide the detailed function and the computation case by case for the threshold $\hat{\eta}_{th}$ and $r_2^{**}(\eta)$ ($\hat{\eta}_{th} < \eta \leq 1$).

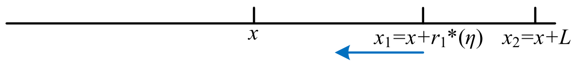


Fig. 6. Illustration of the user's best same-sided reporting of x_1 and x_2 as η increases.

the user's equilibrium reporting strategy on the same side are $(x_1 = x \pm r_1^*(\eta), x_2 = x \pm L)$. *As illustrated in Fig. 6, the smaller equilibrium reporting distance* $r_1^*(\eta)$ *decreases with* η *as follows.*

$$r_1^*(\eta) = \begin{cases} \min\left(r_1^\dagger(\eta), L\right), \\ \text{if } 0 < \eta \le \min\left(\dfrac{2f(2\lambda R)}{f(2\lambda R) + \lambda L f'(2\lambda R)}, 1\right), \\ 0, \text{ otherwise,} \end{cases} \quad (13)$$

where $r_1^\dagger(\eta)$ *is the unique solution to*

$$\left(1 - \frac{\eta}{2}\right) f(\lambda(2R - r_1)) - \lambda\left((1 - \frac{\eta}{2})r_1 + \frac{\eta L}{2}\right) f'(\lambda(2R - r_1)) = 0. \quad (14)$$

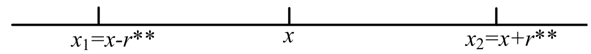
When the adversary is likely to access both geo-data, reporting two locations on the same side of x will lead to the adversary's mean inference with error $\frac{r_1^*(\eta)+L}{2}$. As η increases, the user's expected payoff in (12) is ensured with a greater privacy contributed by $\frac{\eta}{2}L$. Thus the user will reduce $r_1^*(\eta)$ to achieve a greater service gain.

3) Summary of User's Equilibrium Reporting Strategy for $0 \le \eta \le 1$: Comparing the user's expected payoffs when optimally reporting (x_1, x_2) on the same or two opposite sides, we can finalize the user's equilibrium reporting strategy. Before this, we first define a threshold η_{th} as the unique solution to (15), as shown at the bottom of the page, with r^{**} in (11) and $r_1^*(\eta)$ in (13).

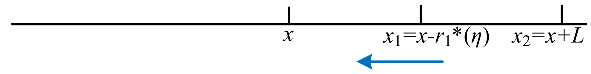
Theorem 1 (User's Equilibrium Reporting Strategy for $0 \le \eta \le 1$*): The user's equilibrium reporting strategy of* (x_1^{BNE}, x_2^{BNE}) *has a threshold structure and depends on* η *as follows (see Fig. 7 for an illustration):*

- *If the adversary has a small probability to access both geo-data, i.e.,* $0 \le \eta \le \eta_{th}$ *with* η_{th} *as the unique solution to (15), the user will strategically report symmetric geo-data* $(x_1^{BNE} = x \mp r^{**}, x_2^{BNE} = x \pm r^{**})$ *on two opposite sides of the real location* x*, where* r^{**} *is given in (11).*
- *If the adversary has a large probability to access both geo-data, i.e.,* $\eta_{th} < \eta \le 1$*, the user will report* $(x_1^{BNE} = x \pm r_1^*(\eta), x_2^{BNE} = x \pm L)$ *on the same side of* x*, where* $r_1^*(\eta)$ *is given in (13).*

Here, η_{th} is the threshold for the user to change from the two-sided reporting strategy focusing on the service gain to the same-sided reporting focusing on the privacy gain. When $0 \le \eta \le \eta_{th}$, the user keeps reporting two symmetric dummy locations on two sides of the real location; Once η exceeds η_{th}, the adversary has a high probability η to access both geo-data



(a) $0 \le \eta \le \eta_{th}$: the user keeps reporting two symmetric locations on two sides of the real location.



(b) $\eta_{th} < \eta \le 1$: the user changes to report on the same side of the real location. As η increases to exceed η_{th}, x_2 jumps to $x + L$, and x_1 moves closer to x.

Fig. 7. Illustration of user's threshold-based reporting strategy at the BNE as η increases from 0 to 1: from two-sided symmetric location reporting in Fig. 7(a) to the same-sided reporting once η exceeds threshold η_{th} in Fig. 7(b).

and use the mean location inference. Hence the user changes to the same-sided reporting strategy by safely reporting a closer location x_1 to achieve greater service gain and a farther location x_2 to achieve greater privacy gain respectively. Fig. 7 illustrates the threshold-based reporting strategy at the BNE.

Corollary 1: The threshold η_{th}*, which is the unique solution to (15), decreases with the PoI density* λ*.*

With a larger λ, the threshold η_{th} is smaller and the user will change his strategy from the two-sided to same-sided reporting earlier. When PoIs are densely located with a larger λ, the user does not worry much about the service gain as it is easier to locate at least one PoI. Instead, he is more likely to choose the same-sided reporting to improve the privacy gain.

Corollary 2: The user's expected payoff $U^{BNE} = U(x_1^{BNE}, x_2^{BNE}|x)$ *at the BNE from (7) decreases with* η *when* $0 \le \eta \le \eta_{th}$ *and increases with* η *when* $\eta_{th} < \eta \le 1$.

Intuitively, when the adversary has a small probability to access both geo-data ($0 \le \eta \le \eta_{th}$), as η increases, the two-sided reported locations lead to a smaller privacy gain. Here the adversary is increasingly likely to infer a more accurate user location. When the adversary is likely to access both geo-data ($\eta_{th} < \eta \le 1$), the user strategically changes to a same-sided reporting strategy. A greater access probability of two geo-data then misleads the adversary to infer the middle point of the same-sided x_1 and x_2 more, which in fact helps the user achieve a greater privacy gain.

C. Comparison With Single-Query Benchmark With $k = 1$

One may wonder how much the user gains from using $k = 2$ queries, comparing with the traditional use of only one query with $k = 1$. In this subsection we analyze the traditional one-query benchmark approach (e.g., [37], [38]) and compare it with our dual-query approach in Theorem 1.

To be fair, similar to the dual-query privacy model in (5), we still assume the user is risk-averse and consider a worst-case privacy gain for $k = 1$. That is, in the single-query case, the user's privacy gain is independent of the adversary's access probability η. Mathematically, the user's expected payoff with the real location x and the only reported

$$\begin{cases} \left((1 - \frac{\eta}{2})r_1^*(\eta) + \frac{\eta}{2}L\right) f(\lambda(2R - r_1^*(\eta))) = (1 - \eta)\, r^{**} f(2\lambda R), & \text{if } 0 \le L \le R, \\ \left((1 - \frac{\eta}{2})r_1^*(\eta) + \frac{\eta}{2}L\right) f(\lambda(2R - r_1^*(\eta))) = (1 - \eta)\, r^{**} f(\lambda(4R - 2r^{**})), & \text{if } L > R. \end{cases} \quad (15)$$

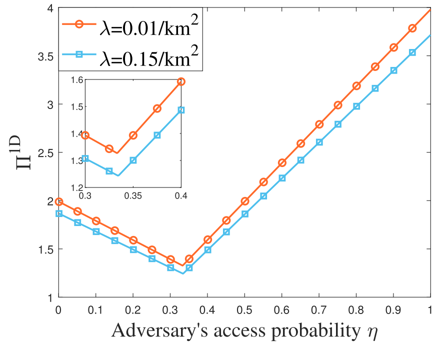


Fig. 8. Ratio Π^{1D} of user's equilibrium expected payoffs for $k = 2$ and $k = 1$. The parameters are set to $R = 1$ km, $L = 4$ km, $\lambda \in \{0.01, 0.15\}/\text{km}^2$. The user's service gain function in Definition 1 takes the specific bounded form of $f(\bar{N}) = 1 - e^{-\bar{N}}$ in this and later simulations.

location x_i of distance r away is

$$\begin{aligned} U^0(x_i|x) &= d\left(\hat{x}_i^{BNE}, x\right) \times f(\lambda S(x_i|x)) \\ &= r \times f(\lambda(2R - r)), \end{aligned} \tag{16}$$

where $S(x_i|x)$ is the service coverage area brought by the single query x_i.

Proposition 6: For the single-query with $k = 1$, the user's equilibrium reporting distance is

$$r^* = \min\left(r^0, L\right), \tag{17}$$

where $r^0 \in (0, 2R)$ is the unique solution to

$$f(\lambda(2R - r)) - \lambda r f'(\lambda(2R - r)) = 0. \tag{18}$$

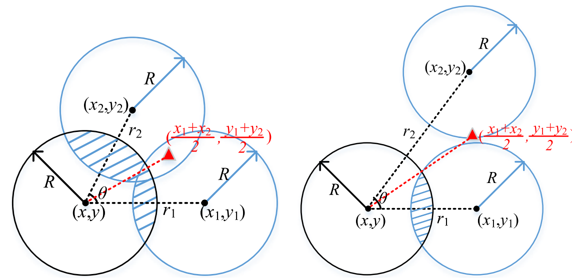
$k = 2$ improves the user's expected payoff than $k = 1$.

In this 1D scenario, to better illustrate the payoff improvement with one more query, we compute Π^{1D} in (19) as the ratio of the user's equilibrium payoffs for $k = 2$ and $k = 1$:

$$\Pi^{1D}(\eta) = \begin{cases} \dfrac{(1-\eta) r^{**} f(2\lambda R)}{r^* f(\lambda(2R - r^*))}, \\ \text{if } 0 \le \eta \le \eta_{th} \text{ and } 0 \le L \le R, \\ \dfrac{(1-\eta) r^{**} f(\lambda(4R - 2r^{**}))}{r^* f(\lambda(2R - r^*))}, \\ \text{if } 0 \le \eta \le \eta_{th} \text{ and } L > R, \\ \dfrac{\left((1 - \frac{\eta}{2}) r_1^*(\eta) + \frac{\eta}{2} L\right) f(\lambda(2R - r_1^*(\eta)))}{r^* f(\lambda(2R - r^*))}, \\ \text{if } \eta_{th} < \eta \le 1, \end{cases} \tag{19}$$

where r^{**} is given in (11), $r_1^*(\eta)$ in (13), η_{th} as the unique solution to (15), and r^* in (17).

In Fig. 8, we plot the ratio $\Pi^{1D}(\eta)$ in (19), which is much greater than 1. Note that Π^{1D} first decreases and then increases with η, similarly as the monotonicity of U^{BNE} in Corollary 2. This is because the user's single-query payoff at the BNE from (16) is a constant independent of the adversary's access probability η. The performance advantage of $k = 2$ over $k = 1$ is intuitive. As long as the user reports with two same queries, the adversary cannot benefit from accessing two geo-data despite its information about the user's strategy, which ensures that $k = 2$ can at least achieve the same performance as $k = 1$. Later in Section V-C, we will further use real dataset to demonstrate the improvement in 2D scenario.



(a) Utilize both (x_1, y_1) and (x_2, y_2) for positive service coverages.

(b) Utilize only (x_1, y_1) for the service gain and (x_2, y_2) for the privacy gain.

Fig. 9. Illustration of the user's reported locations (x_1, y_1) and (x_2, y_2). The reporting distance between (x_i, y_i) and (x, y) is r_i, $i \in \{1, 2\}$, and the smaller angle with sides r_1 and r_2 is defined as θ. When accessing both geo-data, the adversary infers the user's real location based on the mean point $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$ in red.

V. Extension of User's Equilibrium Location Reporting to 2D Ground Plane

In this section, we extend our analysis of the user's reporting for $k = 2$ from a 1D line to a two-dimensional (2D) ground plane. The key difference from Section IV is that here the user has more freedom to decide not only the reporting distances of his reported geo-data but also the reporting angles. Though it is difficult to derive closed-form results, we will show that the key structural results in Section IV still hold here.

Intuitively, when deciding (x_1, y_1) and (x_2, y_2) jointly for LBS, the user may achieve positive service coverages from both geo-data as shown in Fig. 9(a), or use one geo-data (e.g., (x_1, y_1)) to obtain the positive service coverage and the other (x_2, y_2) to achieve the privacy gain as shown in Fig. 9(b). In the latter case, (x_2, y_2) is purposely placed far away so that the adversary's best mean inference $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$ in Proposition 2 is away from the user's real location (x, y).

Without loss of generality, given reporting distances r_1 and r_2 defined in (3), we place (x, y) at the origin $(0, 0)$ on the ground plane, and let $\theta = \langle (x_1, y_1), (x, y), (x_2, y_2) \rangle \in [0, \pi]$ denote the **reporting angle** with sides r_1 and r_2 as shown in Fig. 9. Essentially, θ tells the relative difference between the two dummy locations' directions from the real location. Based on trigonometry, we can uniquely specify the user's reporting strategy through three decision variables (r_1, r_2, θ):[3]

$$\begin{aligned} (x_1, y_1) &= (r_1, 0), \\ (x_2, y_2) &= (r_2 \cos\theta, r_2 \sin\theta). \end{aligned}$$

The total service coverage area $S((x_1, y_1), (x_2, y_2))$ in (4) now depends on r_1, r_2 and θ, and we can rewrite it as $S(r_1, r_2, \theta)$ to give the user's expected payoff function.

Lemma 2: When considering the 2D case, given the adversary's best inference strategy in Stage II (see Propositions 1

[3]Note that when we place (x, y) at the origin and (x_1, y_1) at the horizontal axis, (r_1, r_2, θ) can describe (x_1, y_1) and (x_2, y_2) uniquely. Our focus is the relative positional relationship among the real location and two reported locations rather than the specific direction of the strategic reporting.

and 2), the user's expected payoff function in (6) can be equivalently rewritten as

$$
\begin{aligned}
&U(r_1, r_2, \theta) \\
&= \left((1-\eta) \min_{i=1,2} r_i + \eta \cdot d\left((r_1, 0), (r_2\cos\theta, r_2\sin\theta)\right) \right) \\
&\quad \times f\left(\lambda S(r_1, r_2, \theta)\right). \qquad (20)
\end{aligned}
$$

A. Bayesian Equilibrium Analysis for Special Case of $\eta = 0$

We first consider a special case of $\eta = 0$, i.e., the adversary can only access one of the user's reported locations. In this case, the user does not worry about the adversary's mean location inference strategy in Proposition 2, and the adversary will just regard its only known reported location as the user's real location (see Proposition 1). The user's expected payoff can be simplified from (20) to

$$U(r_1, r_2) = \left(\min_{i=1,2} r_i \right) f\left(\lambda S(r_1, r_2, \theta)\right). \qquad (21)$$

For any $r_1, r_2 \geq 0$, we first present Lemma 3 to show the optimal direction of the strategic reporting.

Lemma 3: Considering the 2D case, if the adversary can only access one of the reported locations with $\eta = 0$, for any reporting distances $r_1, r_2 \geq 0$, the user will strategically report locations (x_1, y_1) and (x_2, y_2) to two opposite directions of the real location at the BNE, i.e., $\theta^{BNE} = \pi$.

The intuition is that to maximize the total service coverage, the user should report two dummy locations to two opposite directions (e.g., on west and east sides) of (x, y) to minimize the overlap between the service coverages brought by two queries. This result is consistent with Lemma 1 in the 1D scenario. Without loss of generality, we assume that $r_1 \leq r_2$. Then the user's expected payoff maximization problem is simplified from (21) to

$$\max_{(r_1, r_2): 0 \leq r_1 \leq r_2} U(r_1, r_2) = r_1 f(\lambda S(r_1, r_2)). \qquad (22)$$

By solving the first-order condition of the concave objective function in (22), we give the equilibrium reporting distances.

Proposition 7: Considering the 2D case, if the adversary can only access one of the reported locations with $\eta = 0$, the user will strategically report with the same distances $r_1^{BNE} = r_2^{BNE} = \bar{r}^{**} = \min(\bar{r}^{\dagger\dagger}, L)$ to two opposite directions of (x, y), where $\bar{r}^{\dagger\dagger} \in (R, 2R)$ is the unique solution to

$$f(\lambda S(r)) - 2\lambda r\sqrt{4R^2 - r^2} f'(\lambda S(r)) = 0,$$

where $S(r)$ is the service coverage brought by $r_1 = r_2 = r$. The user's reporting distance $\bar{r}^{**}$ at the BNE increases with PoI density λ and service searching range R.

Given $\eta = 0$, the adversary cannot average over the two reported locations, hence the user can perform two-sided symmetric strategic reporting (with respect to the real location). As λ or R increases, there are more PoIs within the service coverage area for a greater service gain. Then the user will focus on improving his privacy gain by reporting further away locations.

B. Bayesian Equilibrium Analysis for General $0 \leq \eta \leq 1$

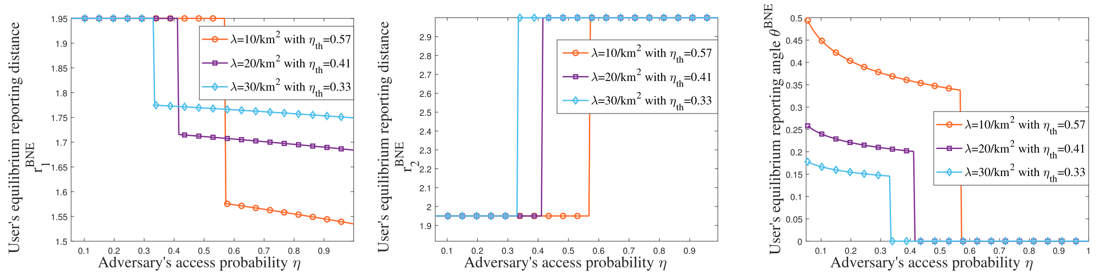
Recall that Theorem 1 explicitly presents the user's equilibrium reporting strategy in the 1D scenario, which is threshold-based from a two-sided reporting to a same-sided reporting. However, given $0 \leq \eta \leq 1$, the total service coverage $S(r_1, r_2, \theta)$ in the 2D scenario changes from case to case and in general is an irregular shape without a closed-form expression (e.g., see shaded area in Fig. 9(a)). Hence it is difficult to analytically solve the user's equilibrium strategy $(r_1^{BNE}, r_2^{BNE}, \theta^{BNE})$. Instead, we will use extensive numerical studies to optimize (20) and identify the key insights, as well as the performance of the proposed approach.

In Fig. 10, we solve and present the user's equilibrium strategy $(r_1^{BNE}, r_2^{BNE}, \theta^{BNE})$ in terms of η and the PoI density $\lambda \in \{10, 20, 30\}/\text{km}^2$. Similar to Theorem 1 for the 1D scenario, here the user's reporting strategy also follows a threshold structure: as η increases, there exists a threshold η_{th} such that the user chooses a different-sided reporting strategy ($0 < \theta^{BNE} < \pi$) with a service advantage when $\eta < \eta_{th}$, while choosing the same-sided reporting strategy ($\theta^{BNE} = 0$) with a privacy advantage when $\eta > \eta_{th}$. We can also observe that the threshold η_{th} decreases with λ, which is consistent with Corollary 1 for the 1D scenario.

To help readers better understand Fig. 10, we present Fig. 11 to summarize our results from Fig. 10, which more clearly shows the user's equilibrium strategy in threshold type.

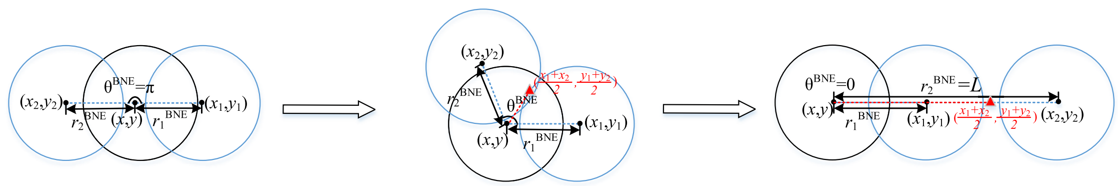
- $\eta = 0$: In Fig. 11(a), (x_1, y_1) and (x_2, y_2) are to the two opposite directions of real location (x, y) with $\theta^{BNE} = \pi$ and $r_1^{BNE} = r_2^{BNE} = \bar{r}^{**}$, as proved in Proposition 7.
- $0 < \eta < \eta_{th}$: In Fig. 11(b), (x_1, y_1) and (x_2, y_2) are with the same reporting distances $r_1^{BNE} = r_2^{BNE}$ which is of the same value as that in Fig. 10(b)-(c). Here (x_1, y_1) and (x_2, y_2) are on two different directions with the angle θ^{BNE} smaller than π and decreasing in η (see Fig. 10(a)), such that the searching disks centered at (x_1, y_1) and (x_2, y_2) overlap more with each other. This keeps the adversary's mean inference of the two reported geo-data away from the real location (x, y), but it yields a smaller total service coverage area as η increases.
- $\eta_{th} \leq \eta \leq 1$: In Fig. 11(c), once η exceeds the threshold η_{th}, θ^{BNE} decreases to 0 (see Fig. 10(a)) to keep (x_1, y_1) and (x_2, y_2) in the same direction of (x, y). The smaller reporting distance r_1^{BNE} suddenly drops (see Fig. 10(b)) and (x_1, y_1) mainly contributes to the service gain. As η further increases, $r_1^{BNE}(\eta)$ decreases and (x_1, y_1) moves closer to (x, y) for more service gain. The larger reporting distance r_2^{BNE} remains as the maximum L (see Fig. 10(c)), to mislead the adversary's mean inference strategy.

In Fig. 12, the user's equilibrium expected payoff U^{BNE} first decreases with $\eta < \eta_{th}$ under the different-sided reporting strategy, and then surprisingly increases with $\eta \geq \eta_{th}$ under the same-sided reporting strategy. This is consistent with our observation in Corollary 2 under the 1D scenario. When $\eta \geq \eta_{th}$, the user's strategic same-sided reporting misleads the adversary to infer the middle point of (x_1, y_1) and (x_2, y_2) more. Then the adversary may not gain from the increasing



(a) User's equilibrium reporting distance r_1^{BNE} on platform 1 versus η. (b) User's equilibrium reporting distance r_2^{BNE} on platform 2 versus η. (c) User's equilibrium reporting angle θ^{BNE} between two geo-data versus η.

Fig. 10. User's equilibrium reporting decisions of the two dummy locations versus η and λ. Here we set searching radius $R = 1$ km, $\lambda \in \{10, 20, 30\}$/km^2, $L = 3$ km. The reporting strategy is of threshold-type and the threshold η_{th} is 0.57 for $\lambda = 10$/km^2, 0.41 for $\lambda = 20$/km^2, and 0.33 for $\lambda = 30$/km^2.



(a) η=0: $\theta^{BNE}=\pi$ and $r_1^{BNE}=r_2^{BNE}$. (b) $0<\eta<\eta_{th}$: $\theta^{BNE}<\pi$ and decreases with η. (c) $\eta_{th}\le\eta\le1$: $\theta^{BNE}=0$, $r_2^{BNE}=L$, and r_1^{BNE} decreases with η.

Fig. 11. Illustration of user's threshold-based reporting strategy of two dummy locations at the BNE as the adversary's probability η of accessing both geo-data increases from 0 to 1, where the user changes from different-sided location reporting to the same-sided reporting once η exceeds threshold η_{th}.

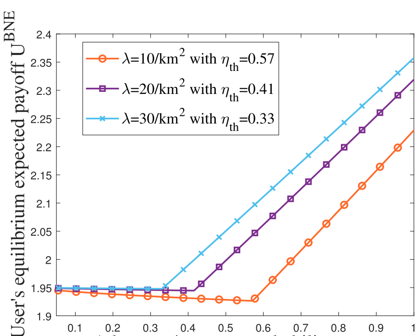


Fig. 12. User's equilibrium expected payoff U^{BNE} versus η. The parameter setting is $R = 1$ km, $\lambda \in \{10, 20, 30\}$/km^2, $L = 3$ km.

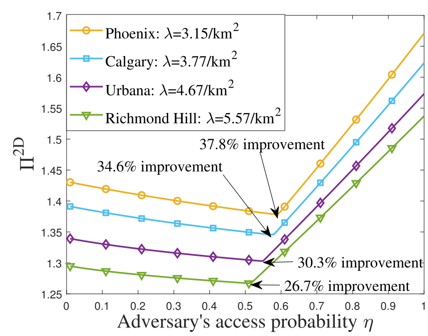


Fig. 13. Ratio Π^{2D} of user's equilibrium expected payoffs under $k = 2$ and $k = 1$. The parameter setting is $R = 1$ km, $\lambda \in \{3.15, 3.77, 4.67, 5.57\}$/km^2, $L = 3$ km.

η to access more geo-data. Only when the adversary happens to access only geo-data (x_1, y_1), it has a closer prediction of the real location as compared to the mean inference from both (x_1, y_1) and (x_2, y_2). Fig. 12 also shows that the user's payoff U^{BNE} increases as the PoI density λ increases, as a greater λ better ensures the service gain and the user can sacrifices more service flexibility for privacy gain.

C. Comparison With Single-Query Benchmark With k = 1

Similar to Section IV-C, here we define Π^{2D} as the ratio of the user's payoffs at the BNE for $k = 2$ and $k = 1$ and express it in (23), as shown at the bottom of the page, for the 2D scenario, where $S(\cdot)$ and $S^0(\cdot)$ are the total service coverage brought by $k = 2$ and $k = 1$, respectively. To evaluate the advantage of the dual-query approach with $k = 2$ over the traditional single-query approach with $k = 1$, we use real dataset [39] from a popular LBS app, Yelp, where users can search for nearby restaurants within a regular walking distance $R = 1$ km. Yelp shares a public dataset covering dozens of cities in Europe and North America, within which the business file contains the yelp's business venues (PoIs) with attributes such as "type", "name" and "category". We focus on the search of restaurants in four cities, Phoenix, Calgary, Urbana, and Richmond Hill, where the average densities of restaurants are $\lambda \in \{3.15, 3.77, 4.67, 5.57\}$/km^2 according to [39].

Figure 13 presents the ratio Π^{2D} in (23) in the 2D scenario. The ratio is significantly larger than 1, which tells that adding one more strategic query in LBS can significantly increase the user's payoff. The ratio $\Pi^{2D}(\eta)$ has the similar monotonicity

$$\Pi^{2D}(\eta) = \frac{\left((1-\eta)r_1^{BNE} + \eta \cdot d\left((r_1^{BNE}, 0), (r_2^{BNE}\cos\theta^{BNE}, r_2^{BNE}\sin\theta^{BNE})\right)\right) f\left(\lambda S(r_1^{BNE}, r_2^{BNE}, \theta^{BNE})\right)}{\bar{r}^* f\left(\lambda S^0(\bar{r}^*)\right)}. \quad (23)$$

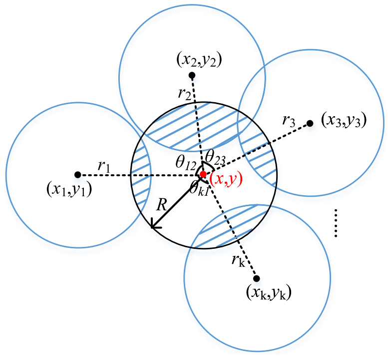
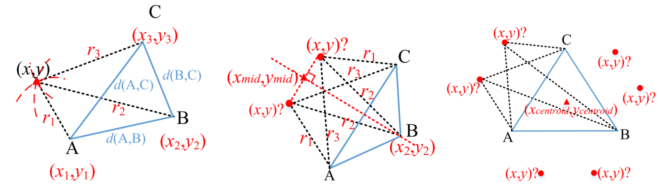


Fig. 14. User's location reporting with k queries.



(a) Given $d(A,B) \neq d(A,C)$, $d(A,B) \neq d(B,C)$ and $d(B,C) \neq d(A,C)$.

(b) Given only $d(A,B) = d(B,C)$ in the isosceles triangle ABC: two possible real locations.

(c) Given $d(A,B) = d(A,C) = d(B,C)$ on the equilateral triangle ABC: six possible real locations.

Fig. 15. Illustration of the adversary's best inference of the user's real location (x, y) from geo-data A, B, C.

property in η as $U^{BNE}(\eta)$ in Fig. 12, i.e., $\Pi^{2D}(\eta)$ decreases with η when $\eta \leq \eta_{th}$ and increases with η when $\eta > \eta_{th}$. In Richmond Hill, where the restaurants are most densely located among the four cities with $\lambda = 5.57/\text{km}^2$, one more query with $k = 2$ can improve the payoff by at least 26.7%.

We can also observe that when the PoI density λ is small (as the Phoenix city in Fig. 13), the ratio Π^{2D} is even larger comparing with other denser cases. Intuitively, the single-query approach has to report the only dummy location close to the real location with less service flexibility, while the dual-query approach gives more freedom for the user to misreport the other dummy location further for privacy gain without sacrificing the service coverage.

VI. Model Extension: General k-Query

In this section, we extend our model and analysis from $k = 1, 2$ to a more general k. Similar to the representative case of $k = 2$ in Section V, we generalize the user's strategy definition with k dummy locations. As illustrated in Fig. 14, the user reports k dummy locations $(x_1, y_1), (x_2, y_2), \cdots, (x_k, y_k)$ on the 2D ground plane. His reporting strategy consists of the reporting distances $(r_1, r_2, \cdots, r_k)$ and the pairwise reporting angles $\theta_{12}, \theta_{23}, \cdots, \theta_{k-1,k}, \theta_{k,1}$ with $\left[\theta_{12} + \theta_{23} + \cdots + \theta_{k-1,k} + \theta_{k1} = 2\pi\right]$.

Similar to Section V, it is difficult to derive analytical solutions for jointly optimizing the user's k reporting distances and k reporting angles, by considering the adversary's best inference responses. In the following, we will consider the worst case that the adversary can access all k geo-data for analytically designing the user's robust strategy, and then compare its performance to that of the k-anonymity mechanisms (e.g., [5], [6]) which only work well under a large number k.

A. User's Robust Reporting Strategy Against the Adversary in the Worst Case

In this subsection, we discuss the worst case when the adversary can access all the user's reported geo-data. Note that we want to provide an alternative to the k-anonymity mechanisms and help the user gain location privacy from service flexibility even for the case of a small number k. Thus, for ease of exposition, we consider the typical multi-query case of $k = 3$ in this subsection. In a similar way, we can analyze the robust mechanism for k more than 3, yet the analysis is on more numerical optimization there.

Suppose the user reports dummy locations A, B and C with mutual distances $d(A, B)$, $d(A, C)$ and $d(B, C)$ on the 2D ground plane. The adversary has full knowledge of the user's payoff function to infer the user's reporting distances (r_1, r_2, r_3) from the three geo-data (x_1, y_1), (x_2, y_2) and (x_3, y_3) to the real location (x, y), respectively, as well as the pairwise angles $(\theta_{12}, \theta_{23}, \theta_{31})$. Note that even in this worst case for the user, the adversary still cannot tell the reporting distance for any dummy location (e.g., point A), which may be r_1, r_2 or r_3. In other words, point A (B or C) can be any of the three geo-data (x_1, y_1), (x_2, y_2) and (x_3, y_3). Before analyzing the user's robust strategy, we first analyze the adversary's best inference attack, depending on the relationships between the three mutual distances $d(A, B)$, $d(A, C)$ and $d(B, C)$ in the following three subcases.

In the first subcase, as illustrated in Fig. 15(a), if any two of the three distances $d(A, B)$, $d(A, C)$ and $d(B, C)$ are found unequal (or equivalently, the triangle ABC is not isosceles), our following analysis shows that the adversary can infer the user's real location (x, y) exactly. Consider an illustrative example in Fig. 15(a), where geo-data A, B, C happen to be the three geo-data (x_1, y_1), (x_2, y_2) and (x_3, y_3), respectively, with reporting distances (r_1, r_2, r_3) and angles $(\theta_{12}, \theta_{23}, \theta_{31})$ from (x, y). Without knowing such connection, the adversary can calculate the mutual distances of the three geo-data, i.e., $d((x_1, y_1), (x_2, y_2))$, $d((x_1, y_1), (x_3, y_3))$ and $d((x_2, y_2), (x_3, y_3))$, given by

$$d((x_i, y_i), (x_j, y_j)) = r_i^2 + r_j^2 - 2r_i r_j \cos\theta_{ij}.$$

Once $d(A, B) = d((x_1, y_1), (x_2, y_2))$ and $d(A, C) = d((x_1, y_1), (x_3, y_3))$, the adversary can recognize point A as (x_1, y_1) with distance r_1 away from (x, y), point B as (x_2, y_2) with distance r_2, and point C as (x_3, y_3) with distance r_3. Finally, the adversary perfectly identifies (x, y) from points A, B and C.

In the second subcase, as shown in Fig. 15(b), suppose only two sides AB and BC of the isosceles triangle ABC are of the same distances (i.e., $d(A, B) = d(B, C)$). Due to this geometric symmetry, the adversary only recognizes point B as (x_2, y_2) with distance r_2 away from (x, y), but is not sure if point A is (x_1, y_1) with distance r_1 or (x_3, y_3) with distance r_3. Thus, there are two candidates for (x, y) in Fig. 15(b), and the adversary infers their mean point (x_{mid}, y_{mid}) as (x, y) to minimize the inference error.

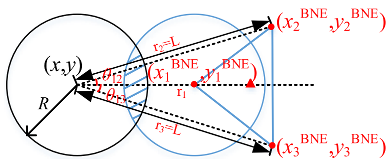


Fig. 16. User's equilibrium strategy for $k = 3$ of equilateral triangle ABC with corners (x_1^{BNE}, y_1^{BNE}), (x_2^{BNE}, y_2^{BNE}), (x_3^{BNE}, y_3^{BNE}), when the adversary accesses all geo-data. Here, the equilibrium reporting strategies are $r_1 = 1.5$ km, $r_2 = r_3 = 4$ km, and $\theta_{12} = \theta_{13} = 54.8°$.

In the last subcase as illustrated in Fig. 15(c), all the triangle ABC's three sides are of the same distances (i.e., $d(A, B) = d(A, C) = d(B, C)$). Due to the symmetry, the adversary cannot recognize any point (A, B or C) as (x_1, y_1), (x_2, y_2) or (x_3, y_3), and there are a total of six possible locations for (x, y) in Fig. 15(c). In this case, the adversary infers the centroid of the triangle ABC as (x, y) to minimize the inference error.

Given the adversary's best strategy as explained in Fig. 15, we are ready to proactively design the user's best reporting structure for the three reported geo-data in the two-stage Bayesian game.

Proposition 8: Given the adversary can access all geo-data in the worst case, at the Bayesian equilibrium, the user should report $\big((x_1^{BNE}, y_1^{BNE}), (x_2^{BNE}, y_2^{BNE}), (x_3^{BNE}, y_3^{BNE})\big)$ *in a way such that these corner points form an isosceles or even equilateral triangle (as illustrated in Fig. 15(b) or Fig. 15(c)), with at least two sides to be of the same distance.*

As the number of geo-data increases from $k = 2$ to $k = 3$, the user's expected payoff increases, to better use another degree of service flexibility to confuse the adversary. Though the equilateral triangle is a special case of the isosceles triangle, they lead to the user's different strategies to confuse the adversary. Thus it is still difficult to decide isosceles or equilateral triangle ABC for the user, as the former has more freedom to misplace dummy locations further from (x, y), while the latter creates more obfuscation to the adversary. Fig. 16 presents a numerical result of the user's best reporting strategy, where the three geo-data turn out to form an equilateral triangle at the BNE here. In this instance, the user uses only one geo-data (x_1^{BNE}, y_1^{BNE}) to achieve the service gain, and the other two geo-data to achieve the privacy gain. The three geo-data are all on the same right-hand side of (x, y), which is similar to Fig. 11(c) under $k = 2$ and $\eta = 1$. Yet using one extra geo-data for privacy gain enables the user to gain more expected payoff here. The adversary, however, does not benefit from accessing all three geo-data but is confused by the strategic reportings.

B. Performance Improvement Over k-Anonymity Mechanisms

Recall that in the traditional k-anonymity mechanisms [5], [6], [40], the user's real location is obfuscated with another $k-1$ dummy locations to confuse the adversary, which works well only for a large number k. By newly exploring the user's service flexibility, here we compare the user's expected payoffs under our strategic approach with the traditional k-anonymity mechanism.

Table II presents the user's payoffs under the k-anonymity mechanism and our multi-query approach with service flexibility, by varying the dummy location number k. Under the k-anonymity, k dummy locations are indistinguishable and the privacy function is defined as the average inference error as similar to (5), while the service gain is always the maximum as the real location is queried for LBS without flexibility. Tables II shows that reporting with 2 dummy locations in our approach already outperforms the complicated 8-anonymity. We can also observe that the user's payoff improvement between $k = 2$ and $k = 3$ is merely 0.27 in our approach, showing that the payoff improvement becomes less obvious in k with a diminishing return with more queries. In other words, we may not need to bother the user to design/report a greater number k of dummy locations for our mechanism, as it already performs well for $k = 3$ or even $k = 2$.

TABLE II

USER'S PAYOFF UNDER k-ANONYMITY AND OUR MULTI-QUERY APPROACH (WHEN THE ADVERSARY ACCESSES ALL GEO-DATA). THE PARAMETER SETTING IS $R = 1$ KM, $\lambda = 10/\text{KM}^2$, $L = 4$ KM

k	1	2	3	4	5	6	7	8
k-anonymity	0	0.34	0.69	1.02	1.36	1.68	2.01	2.36
Multi-query	1.54	2.72	2.99					

VII. MODEL EXTENSION: MULTI-QUERY WITH ARBITRARY LOCATION DISTRIBUTION

In Sections III-VI, we assume that the user's real location is uniformly distributed to give the clean solution and insight. In this section, we aim to relax the assumption to a generalized scenario, where the user's location is arbitrarily distributed and the adversary has such background knowledge. For the model and analysis extensions, here we assume the adversary's prior knowledge of the user's location distribution as $\psi(x)$ with $\int_{x\in\mathcal{M}} \psi(x)dx = 1$, where $\mathcal{M}$ is the continuous region of the user's possible locations.

A. Problem Reformulation for $k = 2$

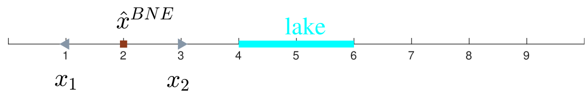
Take the user's dual-query with $k = 2$ for an example, the user's reporting strategy (x_1, x_2) with real location x is now given as a conditional probability $g(x_1, x_2|x)$ with $\int_{x_1}\int_{x_2} g(x_1, x_2|x)dx_1dx_2 = 1$, which guides the mapping probability for any $x, x_1, x_2 \in \mathcal{M}$. Note that the reporting strategy $g(x_1, x_2|x)$ is in probabilistic sense under our generalized model, and when $g(x_1, x_2|x)$ is binary, the reporting strategy is a deterministic one as in Sections IV and V.

Next, we first analyze the adversary's inference $\hat{x}_i^{BNE}$ and $\hat{x}^{BNE}$ when it accesses one or two geo-data as in Section III, and then give the user's payoff function to maximize as in (6). If the adversary can access only one query $x_i \in \{x_1, x_2\}$ as in Section III-A, its optimal inference attack problem with solution $\hat{x}_i^{BNE}$ is given as

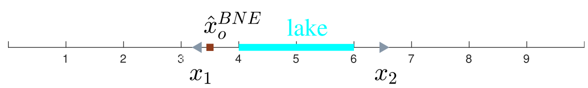
$$\min_{\hat{x}} \int_x \Pr(x|x_i)d(x, \hat{x})dx, \tag{24}$$

while if the adversary can access both geo-data, its inference attack problem with solution $\hat{x}^{BNE}$ is given as

$$\min_{\hat{x}} \int_x \Pr(x|x_1, x_2)d(x, \hat{x})dx, \tag{25}$$

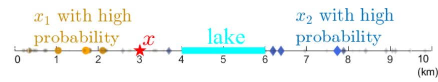


(a) If the two accessed geo-data $x_1 = 1$ km and $x_2 = 3$ km are far away from the lake interval [4 km, 6 km], the adversary's inference attack $\hat{x}^{BNE} = 2$ km still follows the mean inference as in Proposition 2 of Section III.

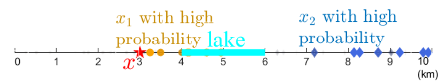


(b) If the two geo-data $x_1 = 3.3$ km and $x_2 = 6.7$ km are close to the lake interval [4 km, 6 km], the adversary's inference attack $\hat{x}^{BNE} = 3.5$ km is no longer the mean of the two accessed geo-data.

Fig. 17. An experiment example to show the adversary's inference attack when it accesses both two geo-data, given that it has prior information about the user's real location ($x \sim U([0, 4 \text{ km}] \cup [6 \text{ km}, 10 \text{ km}])$). The adversary makes the best inference $\hat{x}^{BNE}(x_1, x_2)$ as the solution to (25).



(a) The user's reporting strategy $g(x_1, x_2|x)$ if the adversary has small access probability $\eta = 0.1$ to both geo-data, where the user is likely to report both geo-data (with x_1 in yellow dot and its corresponding x_2 in blue diamond) on the two opposite sides of x.



(b) The user's reporting strategy $g(x_1, x_2|x)$ if the adversary has large access probability $\eta = 0.9$ to both geo-data, where the user is likely to report both geo-data (with x_1 in yellow dot and its corresponding x_2 in blue diamond) on the same side of x.

Fig. 18. An experiment example to show the user's reporting strategy $g(x_1, x_2|x)$ with $k = 2$, given that the adversary has prior information about the user's real location to be evenly distributed except for the lake interval [4 km, 6 km]. The red pentagram denotes the user's real location $x = 3$ km, yellow dots show the reported x_1 and blue diamonds show the corresponding x_2. The size and shade of a dot or diamond tell its appearance frequency or likelihood.

where $\Pr(x|x_i)$ and $\Pr(x|x_1, x_2)$ are posterior probability distributions of the user's real location x, conditional on the accessed queries x_i and (x_1, x_2), respectively. For example, we have

$$\Pr(x|x_1, x_2) = \frac{\Pr(x, x_1, x_2)}{\Pr(x_1, x_2)} = \frac{\psi(x)g(x_1, x_2|x)}{\int_x \psi(x)g(x_1, x_2|x)dx}. \quad (26)$$

Similar to Section II-B, given the real location x and reported locations x_1, x_2, we define the user's privacy gain as $\min_{i=1,2} d\left(\hat{x}_i^{BNE}, x\right)$ if the adversary accesses only one geo-data, and $d\left(\hat{x}^{BNE}, x\right)$ if the adversary accesses both geo-data. We also define the user's service gain similarly as $f(\lambda S)$ with the total service coverage area $S = S(x_1, x_2|x)$. The user's expected payoff is the product of privacy and service gains:

$$\begin{aligned} &U(g(x_1, x_2|x)) \\ &= \int_x \psi(x) \int_{x_1, x_2} g(x_1, x_2|x) f(\lambda S(x_1, x_2|x)) \\ &\times\left((1-\eta)\min_{i=1,2} d\left(\hat{x}_i^{BNE}, x\right) + \eta d\left(\hat{x}^{BNE}, x\right)\right) dx_1 dx_2 dx, \end{aligned} \quad (27)$$

where $\hat{x}_i^{BNE}$ and $\hat{x}^{BNE}$ denote the adversary's optimal inference attacks as the solution to (24) and (25), respectively.

To maximize the user's expected payoff in (27) by designing the reporting function $g(x_1, x_2|x)$, we need to input the embedded parts $d\left(\hat{x}_i^{BNE}, x\right)$ and $d\left(\hat{x}^{BNE}, x\right)$ from the adversary's inference attacks $\hat{x}_i^{BNE}$ and $\hat{x}^{BNE}$ in (24) and (25), both of which are no longer in closed-form. We can numerically solve the payoff maximization problem with payoff in (27) to show similar designs and insights, though the computational complexity scales up with the size of region set $\mathcal{M}$. Next we use an experiment to obtain and explain the results.

Suppose the user's reporting strategy $g(x_1, x_2|x)$ is probabilistic and consider a simple example: the user's real location is evenly distributed in a 1D line [0, 10 km] except for a lake interval (4 km, 6 km) where the user cannot be located. First we examine the adversary's inference attack as in (25) leading to the minimum expected distance to all possible real locations given the accessed geo-data. With an unevenly distributed user location, Fig. 17(a) shows that the adversary still takes the mean inference attack when it accesses both geo-data if both reported x_1 and x_2 are far from the lake, which is similar to Proposition 2 of Section III-B. Yet the adversary will no longer use the mean inference attack if both x_1 and x_2 are close to the lake as shown in Fig. 17(b).

In a probabilistic manner, Fig. 18 further illustrates the user's reporting strategy $g(x_1, x_2|x)$ for different access probability η to both geo-data when the real location is $x = 3$ (denoted by the red pentagram). Fig. 18(a) shows the user's reporting strategy $g(x_1, x_2|x)$ if the adversary has a small access probability to both geo-data, where the user is likely to report both geo-data (with x_1 in yellow dot and its corresponding x_2 in blue diamond) on the two opposite sides of x. The size and shade of a dot or diamond tells its appearance frequency or likelihood. As a lake interval is located on the right-hand side of x, we observe that x_2 mostly in blue diamond is further reported to the right-hand side of the lake. Fig. 18(b) shows the user's reporting strategy $g(x_1, x_2|x)$ if the adversary has a large access probability to both geo-data, where the user is likely to report both geo-data (with x_1 in yellow dot and its corresponding x_2 in blue diamond) on the same right-hand-side of x. Here x_1 near x is to obtain the service gain while x_2 far away is used to confuse the adversary. The simulation result here is consistent with our results in Theorem 1 of Section IV: with high probabilities, the user is likely to report two dummy locations x_1 and x_2 on the two-side of the real location x if the adversary's access probability η is small, while is likely to report locations on the same-side of x if η is large.

B. Comparison With Modified k-Anonymity Mechanisms

Considering the adversary's prior information about the user's location distribution $\psi(x)$, we compare our extended

TABLE III

USER'S PAYOFF UNDER MODIFIED k-ANONYMITY AND THE EXTENDED MULTI-QUERY (WHEN $x \sim U([0, 4\text{ KM}] \cup [6\text{ KM}, 10\text{ KM}])$ AND THE ADVERSARY ACCESSES ALL GEO-DATA). THE PARAMETER SETTING IS $R = 0.5$ KM, $\lambda = 0.5/\text{KM}^2$

k	1	2	3	4	5	6
k-anonymity	0	0.31	0.59	0.71	1.22	1.56
Multi-query	0.51	0.98	1.76			

solution with the modified k-anonymity mechanism with rigid service requirement in [40], i.e., to only choose dummy locations with similar query probabilities. Tables III presents the user's payoffs under the modified k-anonymity mechanism and the multi-query approach, both considering the adversary's prior information. By strategically querying with $k = 2$ dummy locations, our multi-query approach still outperforms the 6-anonymity approach, even if the adversary has the full information about the user's arbitrary location distribution.

C. Extension to Continuous-Time LBSs

Our approach can be extended to continuous-time LBSs with multiple queries over multiple time slots under certain assumptions. Here we utilize Markov chain to model the spatio-temporal correlation in the queries over time. Specifically, we assume the user's location is in a spatial set $\mathcal{M}$ consisting of M seperate regions. The initial location distribution is $\boldsymbol{\psi}_0(x)$ with $x \in \mathcal{M}$ and the $M \times M$ transition matrix is P as the adversary's background information. If the user is at $x^t = i$ at time slot t, the prior location distribution at $t+1$ becomes:

$$\psi_{t+1}(x^{t+1} = j | x^t = i) = p_{ij}. \tag{28}$$

From the perspective of the adversary who can only observe earlier reported locations, its background belief of the user's location distribution is different from the real one in (28). At time slot $t+1$, the location distribution in the adversary's belief is

$$\hat{\psi}_{t+1}(x^{t+1} = j | \boldsymbol{x}_o^t) = \sum_{i \in \mathcal{M}} p_{ij} \Pr(\hat{x}^t = i | \boldsymbol{x}_o^t).$$

where $\boldsymbol{x}_o^t$ denote the adversary's observations of the user's reported geo-data at time slot t. The location distribution $\hat{\psi}_{t+1}(x)$ in the adversary's belief might not be accurate by using its estimated $\hat{x}^t$. The posterior probability, $\Pr(\hat{x}^t = i | \boldsymbol{x}_o^t)$, is dependent on the user's location distribution $\hat{\boldsymbol{\psi}}_t(x)$ in the adversary's belief. For example, if the adversary accesses both geo-data at time slot t, i.e., $\boldsymbol{x}_o^t = (x_1^t, x_2^t)$, then

$$\begin{aligned}\Pr(\hat{x}^t = i | \boldsymbol{x}_o^t) &= \frac{\Pr(\hat{x}^t = i, x_1^t, x_2^t)}{\Pr(x_1^t, x_2^t)} \\ &= \frac{\hat{\psi}_t(\hat{x}^t = i) g(x_1^t, x_2^t | \hat{x}^t = i)}{\sum_x \hat{\psi}_t(x) g(x_1^t, x_2^t | x)},\end{aligned}$$

which takes a similar form as (26). Then the user's privacy preservation problem per time slot can also be formulated as a maximization problem with a payoff function similar to (27) with the real location distribution $\psi^{t+1}(x)$ in (28). The adversary's inference problem can be formulated as a minimization problem similar to (24) or (25), yet with the different location distribution $\hat{\psi}^{t+1}(x)$. The extra issue in the sequential optimization problem over time is that its computational complexity scales up with time.

VIII. CONCLUSION

In this paper, we explored how a user best uses his service flexibility to gain location privacy, by jointly reporting multiple dummy locations for LBS. Compared with the traditional k-anonymity mechanisms, the proposed Bayesian-game approach relieves the user of reporting a great number of dummy locations to the LBS platform. By analyzing the interaction between the user and the adversary, we show that the user's reporting strategy is heavily influenced by the adversary's capability of geo-data access. The user's payoff increases in the number of queries, yet with a diminishing return. Numerical results show that our approach performs well than the k-anonymity mechanism, even if the adversary has the prior information about the user's location distribution, especially for a small number k.

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